

Godwin Praise Tochukwu

AI/ML ENGINEER

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PROFESSIONAL SUMMARY

Machine Learning and AI Engineer with hands-on experience building core ML and DL models, including sentiment analysis systems and medical imaging classification. Strong foundation in Python, data science workflows, model training, evaluation, and explainable AI (XAI). Passionate about applying AI to solve real-world problems, especially in education, healthcare, and automation.

TECHNICAL SKILLS

- Programming: Python, SQL
- Machine Learning: Scikit-learn, Pandas, NumPy
- Deep Learning: TensorFlow, Keras, PyTorch, ONNX (model export & optimization)
- NLP & RAG: Tokenization, TF-IDF, Text Classification, Retrieval-Augmented Generation (RAG), ChromaDB, Sentence-Transformers, LLM API Integration (Groq)
- Computer Vision: CNNs, EfficientNet, Image Preprocessing, Data Augmentation, Grad-CAM
- Backend & APIs: FastAPI, Django REST Framework, PostgreSQL, JWT Authentication, Microservices Architecture, NGINX, Docker
- Frontend: React (Vite), Next.js, Tailwind CSS
- Tools: Git, GitHub, Jupyter Notebook, Google Colab, Docker Compose
- Practices: Model Evaluation, Feature Engineering, Explainable AI (XAI), Out-of-Distribution (OOD) Detection, Version Control, API Integration

PROJECTS

- Chest X-Ray Pneumonia Classification:

Developed a Convolutional Neural Network (CNN) model to classify chest X-ray images as Normal or Pneumonia. Used data augmentation, model training using

TensorFlow/Keras, and evaluation metrics such as accuracy and recall. Applied Grad-CAM for model explainability (XAI) to visualize areas influencing predictions.

- Sentiment Analysis on Text Reviews:

Built an NLP model using TF-IDF and machine learning classifiers to determine sentiment categories (positive, negative, neutral). Performed text preprocessing including tokenization, stopwords removal, and handling class imbalance. Selected the best-performing model based on F1 score.

- Plant Disease Detection System:

Built a full ML pipeline to classify plant diseases using an EfficientNet-B0 model trained on the PlantVillage dataset. Served the model via FastAPI with ONNX for lightweight, fast inference. Designed a two-gate pipeline combining a pretrained MobileNetV2 ONNX model for plant-subject gating with a confidence threshold on the disease classifier to reject out-of-distribution inputs. Targeted free-tier deployment on Render for accessible, low-cost hosting. Structured as part of a broader computer vision roadmap progressing from classification to object detection and segmentation.

- AI-Powered RAG Assistant (NLP / Retrieval-Augmented Generation Project)

Built a fintech-focused conversational assistant using Retrieval-Augmented Generation (RAG), with ChromaDB and sentence-transformers for document retrieval and Groq as the LLM provider. Built the backend with FastAPI and scaffolded a React (Vite) frontend. Generated a synthetic dataset to test and validate assistant responses.

SOFT SKILLS

- Analytical & Critical Thinking
- Problem Solving
- Research-Oriented
- Attention to Detail
- Collaboration & Communication

EDUCATION

Veritas University – B.Sc. Computer Science

Prime Scholars' School, Karu, Abuja